SATA 1D: Two Questions in Depth

1. Why SATA per sub-band gives the same result as SATA whole band
2. Why the ambiguities stop growing with the height of the targets

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Setup. SATA 1D, $N_{rx} = 4$ receive channels on the DPCA condition ($\Delta b_{at} = 7.69$ m), PRF = 2000 Hz, per-channel $\mathrm{PRF}_{\mathrm{ch}} = 500$ Hz, processed band $B_a = 1235$ Hz, cross-track baselines $b_{xt} = 63.7, 27.0, 4.1, 1.7$ m, $r_0 = 766.2$ km, look angle $\theta = 20°$, $\lambda = 0.25$ m, platform velocity $v = 7689$ m/s. Iso-range point targets. *Highest ambiguity*: largest value of the focused image within ± 20 ms of the positions where the copies of a target appear ($x_j/v \pm 0.81$ s and ± 1.62 s), in dB relative to the peak of the ideal (single-channel, full-PRF) reference. All numbers below come from the current sweep (1053 cases) and from the probe scripts listed at the end.

Question	Answer in one paragraph
1. Sub-band = whole band	It is exact for this geometry, not a coincidence. The residual of an elevated iso-range target is a change of the receiver's closest-approach range, $\Delta\rho_i = -b_{xt,i}\, h \cot\theta / r_0$, and along the aperture it varies only as $\Delta\rho_i \cos\beta$, with the azimuth angle $\beta \le 0.6°$: the Doppler dependence is $\beta^2/2 \approx 5 \times 10^{-5}$ of the residual (0.16° at $h = 10$ km, 0.5° at 37 km). Both kernels therefore rotate every time–frequency cell of the target by the same phase. The per-sub-band code moreover re-fits ΔC_0, which is defined at closest approach and cannot differ between bands, so it equals the whole band by construction even when a Doppler-dependent residual is injected. When such a residual is injected, what recovers the reference level is a Doppler-dependent *map* (constant + linear term) in the whole-band kernel; a per-band constant makes the ambiguity worse.
2. Saturation	The reconstruction with channel phase errors φ_i is a unitary mixing of the target's energy between its main lobe and its copies, $c(d) = \frac{1}{N}\sum_i e^{j\varphi_i} e^{-j2\pi di/N}$ with $\sum_d \lvert c(d)\rvert^2 = 1$. The copy at ± 0.81 s cannot exceed $\beta_1 = 1 - \mathrm{PRF}_{\mathrm{ch}}/B_a = 0.595$ (-4.5 dB) of the reference peak. The phases grow linearly with height (2π per 1.1 / 2.6 / 17 / 42 km on the four channels) but enter through $e^{j\varphi}$: beyond one cycle the copies stop growing and fluctuate around -7 dB. Measured -4.9 dB at most over 1053 cases. What does keep growing is the ambiguity relative to the target's *own* peak: $+33$ dB at $h = 4.2$ km, where the main lobe is at -38 dB. The two normalisations answer different questions; the heatmaps use the first.

1 Why SATA per sub-band gives the same result as SATA whole band

1.1 What the two schemes do

A channel line is sampled at $\mathrm{PRF}_{\mathrm{ch}}$, so its spectrum is folded: a target whose true Doppler is f appears at the baseband frequency $f_b = f - m\mathrm{PRF}_{\mathrm{ch}}$ for the fold m that brings it into $[-\mathrm{PRF}_{\mathrm{ch}}/2, \mathrm{PRF}_{\mathrm{ch}}/2)$. SATA works on sub-apertures of 20 samples (40 ms): in each sub-aperture spectrum a target occupies one bin, because its Doppler changes by only $K_a \cdot 40\,\mathrm{ms} = 25$ Hz within the sub-aperture ($K_a = 2v^2/\lambda r_0 = 617$ Hz/s), far less than $\mathrm{PRF}_{\mathrm{ch}}$. Every bin is mapped to the azimuth cell $x = r_0 \tan(\arcsin(\lambda f_b/2v))/v \cdot \mathrm{PRF}_{\mathrm{ch}}$ and multiplied by $e^{j2\pi\,\Delta C_0[x]/\lambda}$.

Whole band labels the bin with f_b, so the fold m of a target is read m image widths ($X = 6.23$ km) away from the target; the map carries copies of every target's block at $0, \pm X$ so that each fold reads its own value. *Per sub-band* k labels the bin with a frequency in $[f_k - \mathrm{PRF_{ch}}/2, f_k + \mathrm{PRF_{ch}}/2)$, $f_k = (-2 + k + \frac{1}{2})\mathrm{PRF_{ch}}$, which moves the images to a one-sided set, and reconstructs output band k from the channels corrected for k. In both schemes the value written in the map is ΔC_0 of the target; in the per-sub-band scheme it is re-fitted over the look angles of band k (`getcoeff_beam`).

1.2 The residual of an elevated target is a constant range offset

Transmitter at $(vt, 0, H)$, receiver i at $(vt - b_{at,i},\, b_{xt,i},\, H)$, target at (x, y, h) with $y(h) = \sqrt{r_0^2 - (H-h)^2}$ so that the monostatic slant range is r_0 for every height (iso-range). The monostatic range history $2\sqrt{r_0^2 + v^2(t - x/v)^2}$ is then the same for all heights, and the whole residual with respect to the flat target at the same slant range sits in the receiver's range:

$$\delta R_i(t) = |\mathbf{p}_{rx,i}(t) - \mathbf{p}(h)| - |\mathbf{p}_{rx,i}(t) - \mathbf{p}(0)|, \qquad \rho_i(h)^2 = r_0^2 - 2\,y(h)\,b_{xt,i} + b_{xt,i}^2, \tag{1}$$

where ρ_i is the receiver's closest-approach range. With $\tau = t - (x + b_{at,i})/v$ and $\tan\beta = v\tau/\rho_i$,

$$\delta R_i(\tau) = \sqrt{\rho_i(h)^2 + v^2\tau^2} - \sqrt{\rho_i(0)^2 + v^2\tau^2} = \Delta\rho_i \cos\beta + O(\Delta\rho_i^2/\rho_i), \tag{2}$$

$$\Delta\rho_i = \rho_i(h) - \rho_i(0) \simeq -\,b_{xt,i}\,\frac{h\cot\theta}{r_0},$$

where β is the azimuth angle under which the receiver sees the target. The constant $\Delta\rho_i$ is the ΔC_0 that SATA corrects ($2\pi\Delta\rho_i/\lambda = -164°$ for $h = 500$ m on channel 0, one cycle per $h_{2\pi,i} = \lambda r_0/(b_{xt,i}\cot\theta) = 1.10, 2.6, 17, 42$ km on the four channels). Its only dependence on the look angle is the factor $\cos\beta$. With Doppler $f = 2v\sin\beta/\lambda$,

$$\delta\varphi_i(f) = \varphi_i\,(1 - \cos\beta) \simeq \varphi_i\,\frac{1}{2}\Big(\frac{\lambda f}{2v}\Big)^2, \qquad \frac{1}{2}\Big(\frac{\lambda f}{2v}\Big)^2 = 5.0\times10^{-5} \text{ at } f = \pm B_a/2. \tag{3}$$

Table 1 checks Eq. (2)–(3) against the code: the closed form reproduces the ΔC_0 of `GetCoeffNu` to the micrometre, and the exact variation of δR_i across the processed band equals $\Delta\rho_i(1 - \cos\beta_{edge})$. Even at $h = 37$ km (the highest case of the sweep) the residual changes by 0.5° across the band, and the four sub-band centres differ from the whole-band value by 0.08–0.76°. There is nothing for a per-sub-band correction to resolve.

h [m]	$\Delta\rho_0$ closed form [mm]	ΔC_0 from `GetCoeffNu` [mm]	φ_0 [deg]	variation across B_a, exact	Eq. (3)	sub-band centres ±750 / ±250 Hz
100	−22.828	−22.828	−32.9	0.0017°	0.0017°	−0.0024° / −0.0003°
500	−113.870	−113.870	−164.0	0.0083°	0.0083°	−0.012° / −0.0014°
1000	−227.073	−227.073	−327.0	0.017°	0.017°	−0.024° / −0.003°
3000	−673.410	−673.410	−969.7	0.049°	0.049°	−0.072° / −0.008°
10000	−2161.102	−2161.102	−3112.0	0.158°	0.157°	−0.231° / −0.026°
37000	−7082.864	−7082.864	−10199.3	0.516°	0.514°	−0.758° / −0.084°

Table 1: Channel 0 ($b_{xt} = 63.7$ m), one target at $x = 0$. The other channels scale with $b_{xt,i}$. The look-angle dependence of the residual is 5×10^{-5} of the residual itself.

1.3 Consequence: both kernels apply the same phase to every cell

Since the value to apply is the same at every Doppler, the two kernels differ only in how they label the bins and where they place the images, and Fig. 1 shows that on the cells that hold the target's energy the applied phase is identical (energy on cells that differ by more than 1°: 0 % for one target, 0.011 % for two targets 2.6 km apart, where the two image sets overlap differently at the block edges). The reconstruction then receives the same corrected channels, up to the image-overlap details, and gives the same result. Over the sweep, |per sub-band −

whole band| is at most 0.32 dB with the targets 200 m apart (543 cases), 0.51 dB at 2.6 km, and reaches 2.5 dB only at 10 km at levels below −48 dB, where both read the sidelobes of the neighbours rather than an ambiguity.

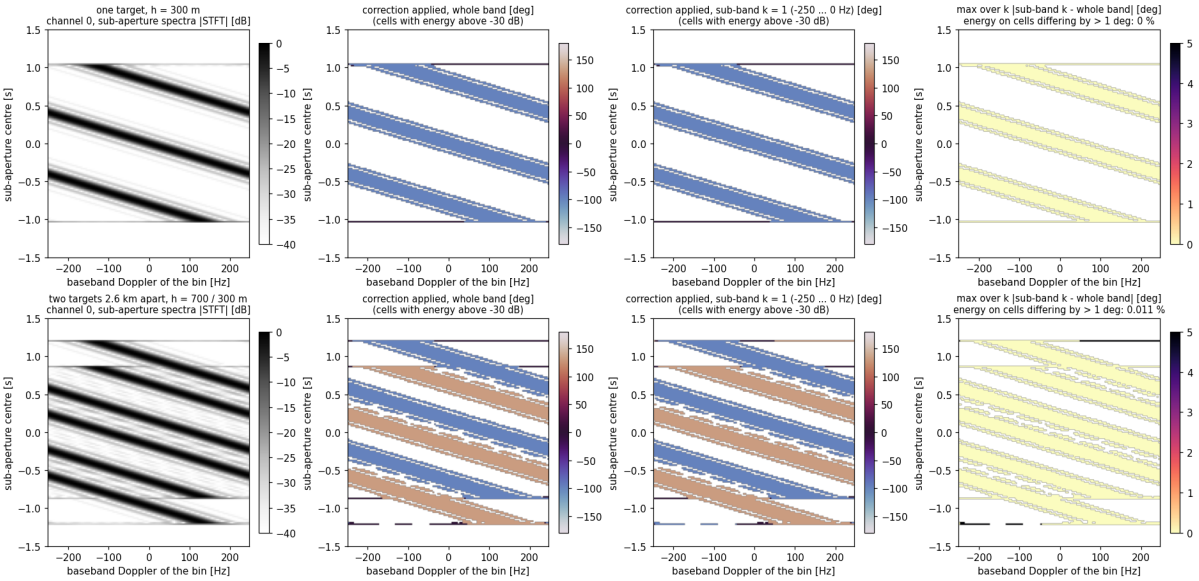


Figure 1: Channel 0. Left: sub-aperture spectra of the channel line (one target at $h = 300$ m; two targets 2.6 km apart at 700 and 300 m). Middle: phase applied by the whole-band kernel and by the kernel of sub-band 1. Right: largest difference over the four sub-band kernels, on the cells within 30 dB of the peak.

1.4 What it would take for the two schemes to differ

A Doppler-dependent residual. To see how each scheme responds to one, the single target at $h = 500$ m was given, on top of its geometric ΔC_0, a delay $\delta t_i = \delta t_0\, b_{xt,i}/b_{xt,0}$ of its bistatic range history in channel i, i.e. the registration residual ΔD_t that a formation with converging tracks would produce, whose phase $2\pi f \delta t_i$ is linear in Doppler. Six corrections were run (Fig. 2, Table 2):

phase of channel 0 at the band edge	0°	11°	22°	44°	89°	main lobe at 89°
no SATA	−7.5	−7.4	−7.4	−7.3	−7.1	−5.8
SATA whole band (current)	−59.7	−37.7	−31.8	−25.7	−19.5	−1.3
SATA per sub-band (current)	−59.7	−37.7	−31.8	−25.7	−19.5	−1.3
per sub-band, residual evaluated at f_k	−59.7	−32.4	−26.3	−20.1	−14.1	−0.5
whole band, Doppler-dependent map (constant + linear)	−59.7	−59.6	−59.5	−59.6	−59.4	−0.3
exact correction	−63.6	−63.6	−63.6	−63.6	−63.6	−0.3

Table 2: Highest ambiguity [dB re. reference peak] with an injected Doppler-dependent residual; last column: main-lobe peak. Reference: −69.2 dB.

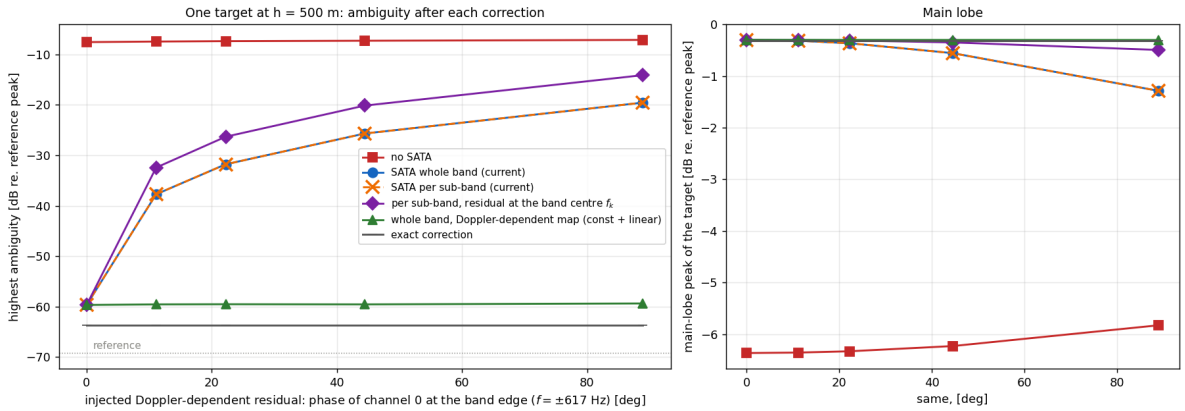


Figure 2: Left: ambiguity after each correction against the size of the injected Doppler-dependent residual. Right: main lobe of the target.

- **The current per-sub-band scheme equals the whole band to the second decimal even here.** It re-fits ΔC_0 over each band's look angles, but ΔC_0 is the constant of the fitted parabola completed to its stationary point, a property of the target's closest approach; the same number comes out for every band. The scheme carries the same quantity as the whole band and cannot differ from it, in this or any geometry.
- **A per-band constant equal to the residual at f_k makes the ambiguity worse** (by 5 dB), although it improves the main lobe. Each sub-aperture cell holds one fold of the target; the band-k value is right for the energy that truly belongs to band k (hence the better main lobe), but the same cell in the same channel line also carries, at other times, the other folds, which the reconstruction has to cancel; a constant that is wrong by $|f_k - f|\,\delta t_i$ on those folds, differently on each channel, spoils the cancellation.
- **A Doppler-dependent map in the whole-band kernel restores the reference level** at every injected size: the map carries a constant and a linear term (ΔC_0 and δt), the images carry the fold-aware constants $\lambda\,\delta t_i\,m\mathrm{PRF}_{\mathrm{ch}}$, and the kernel applies $-2\pi[\Delta C_0 + \lambda\,\delta t\,f_b]/\lambda$ per bin. Because each cell holds a single fold, the whole-band kernel can apply any dependence on Doppler exactly; the sub-band split adds nothing to it.
- SATA is very sensitive to such a residual: 11° at the band edge already costs 22 dB. In this geometry there is none to be sensitive to.

1.5 When is the residual Doppler-dependent?

- **Spaceborne formation, parallel tracks (this study):** Eq. (3), 5×10^{-5} of the residual. The baseline is perpendicular to the velocity, and a perpendicular displacement projects onto the line of sight as $\cos\beta$ over a beam of 1.2°.
- **Formation dynamics:** a cross-track baseline that changes within the synthetic aperture (±1 s) gives a height-dependent registration shift $\Delta D_t \simeq \dot{b}_{xt}\,\Delta y/v^2$ ($\Delta y = h\cot\theta$). For a relative cross-track velocity of 0.3 m/s (helix-type formations) and $h = 500$ m this is 8 µs, 2° at the band edge: a few dB on a residual at −60 dB, invisible next to the 50 dB that SATA removes.
- **Airborne SAR** (the origin of the sub-aperture idea in SATA): beams of tens of degrees, so $1 - \cos\beta$ reaches per cent, and motion errors that change along the track; there the residual depends on the sub-aperture and the correction must follow it.

- **2D processing:** the energy of sub-band k sits in a different range bin (range migration $r_0(1/\cos\beta - 1) \approx 40$ m at the band edge). The phase to apply is the same; what changes with the sub-band is *where* it is applied. That is a placement problem, which a 2D map handles, not a value problem.

Conclusion. In the 1D iso-range geometry with parallel tracks the residual of a target is one number per channel, to 10^{-4} relative. SATA whole band and SATA per sub-band both apply that number to all of the target's energy, so they are the same algorithm. The equality is a statement about the geometry, verified to the micrometre, and the machinery does produce different results when a Doppler-dependent residual is present (Table 2), so it is not a wiring artefact. If a Doppler-dependent correction is ever needed, the change is to the *map* (a ΔD_t / ΔC_1 term next to ΔC_0, with fold-aware images), and the whole-band kernel already has the structure to apply it.

2 Why the ambiguities stop growing with the height

2.1 The reconstruction with phase errors is a unitary mixing

On the DPCA condition, channel i samples the monostatic signal at $t_n + iT$ ($T = 1/\mathrm{PRF}$). At baseband frequency f_b its spectrum is $S_i(f_b) = \frac{1}{N} e^{j2\pi f_b iT} \sum_m S(f_b + m\mathrm{PRF_{ch}})\, e^{j2\pi mi/N}$: a DFT over the N folds. The reconstruction inverts this. A target whose channels carry the extra phases φ_i (here $\varphi_i = 2\pi\Delta\rho_i/\lambda$) comes out as

$$S_{rec}(f_b + k\mathrm{PRF_{ch}}) = \sum_m c(k-m)\, S(f_b + m\mathrm{PRF_{ch}}), \qquad c(d) = \frac{1}{N}\sum_{i=0}^{N-1} e^{j\varphi_i}\, e^{-j2\pi di/N}, \tag{4}$$

$$\sum_d |c(d)|^2 = \frac{1}{N}\sum_i |e^{j\varphi_i}|^2 = 1. \tag{5}$$

$c(0)$ scales the main lobe; $c(d)$ moves the part of the spectrum that survives a shift of $d\mathrm{PRF_{ch}}$ into a focused copy at $d\,\Delta t_{amb}$, $\Delta t_{amb} = \mathrm{PRF_{ch}}/K_a = 0.81$ s. Only the fraction $\beta_{|d|} = (B_a - |d|\mathrm{PRF_{ch}})/B_a$ of the band survives the shift: $\beta_1 = 0.595$ (-4.5 dB), $\beta_2 = 0.19$, $\beta_3 = 0$. Hence three facts that do not depend on the size of the errors:

1. there are at most four copies, at ± 0.81 and ± 1.62 s;
2. the copy at ± 0.81 s cannot exceed $\beta_1 = -4.5$ dB of the reference peak, and reaches it only when all the energy goes into one neighbouring band (φ_i linear in i);
3. the energy the copies gain, the main lobe loses.

The φ_i grow linearly with height, but enter Eq. (4) through $e^{j\varphi_i}$. For $\varphi \ll 1$, $c(d) \simeq \frac{j}{N}\sum_i \varphi_i e^{-j2\pi di/N} \propto h$: the ambiguity grows by 20 dB per decade of height. Once the largest phase passes about 1 rad ($h \approx 175$ m on channel 0) the linearisation fails, and once it passes 2π the phases are effectively random: more height gives different, not larger, values. For random phases $\mathrm{E}|c(d)|^2 = 1/N$, i.e. -6 dB, or -10.5 dB after β_1.

2.2 One target, height 0 to 6 km

Fig. 3 follows one target at $x = 0$ through the four regimes. The prediction from Eq. (4), using nothing but the four phases of Table 1, matches the simulation within 0.26 dB for the copies at ± 0.81 s over the whole range, within 1.3 dB for the main lobe, and $\sum_d |c(d)|^2 = 1$ holds to 10^{-15}. The measured slope between 50 and 200 m is 20.5 dB per decade. The ambiguity crosses -10.5 dB at $h = 330$ m, channel 0 reaches 180° at 550 m, and the largest value, -5.0 dB, occurs at $h = 4.25$ km against the bound of -4.5 dB.

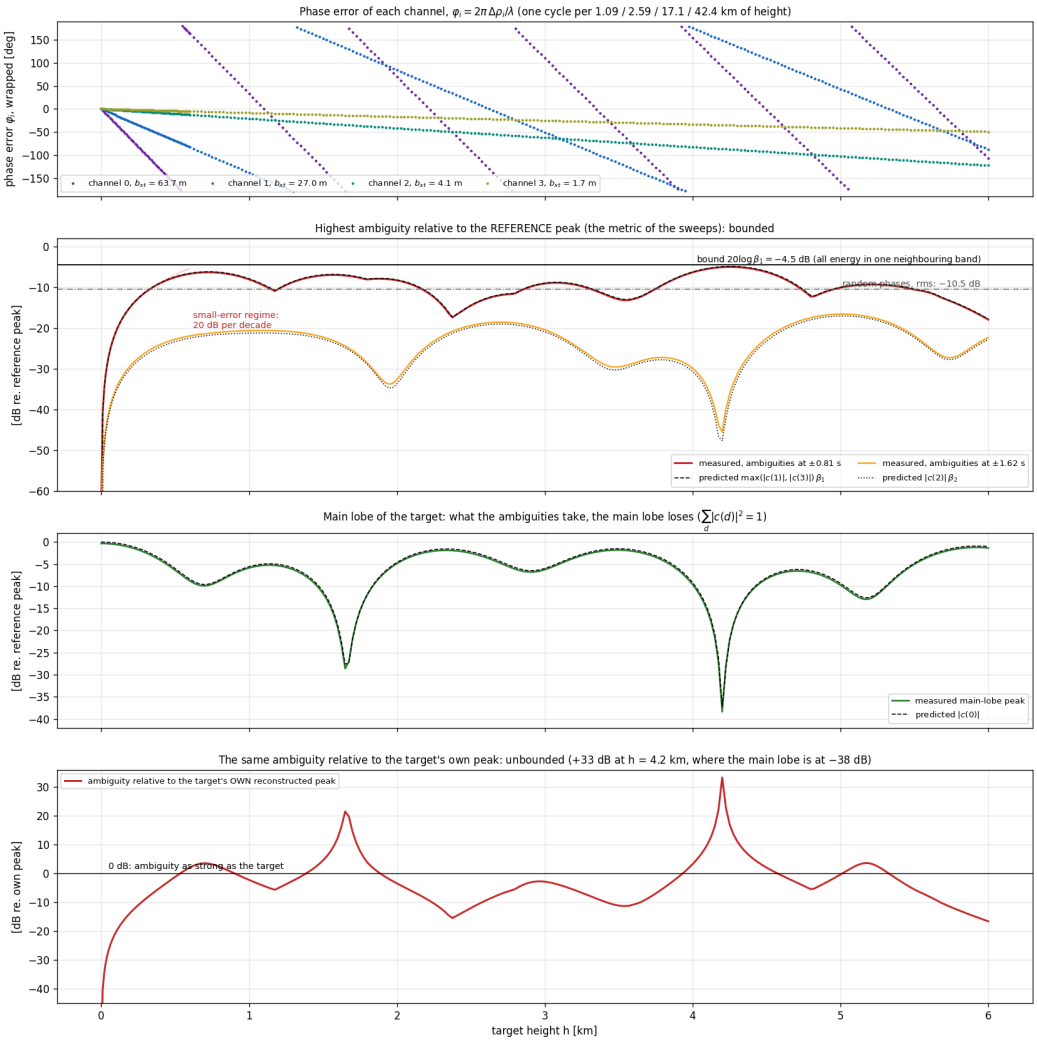


Figure 3: One target, no SATA, height 0–6 km. From top: channel phase errors; highest ambiguity relative to the reference peak, measured and predicted; main lobe; the same ambiguity relative to the target's own reconstructed peak.

2.3 What keeps growing and what does not

The expectation that the ambiguity should keep growing is right for one normalisation and wrong for the other, and both are legitimate:

- **Relative to the reference peak** (the metric of the sweeps and heatmaps, i.e. the absolute strength of the copy in the image): bounded by β_1, by energy conservation. This is the quantity that decides whether the copy is visible next to other scatterers of the scene, and it saturates.
- **Relative to the target's own peak** (the ratio a target-to-ambiguity figure would quote): unbounded, because the main lobe itself collapses when the copies take its energy. In Fig. 3 the main lobe drops to −28 dB at $h = 1.65$ km and to −38 dB at 4.2 km, where the copy is 33 dB *stronger* than the target. The target has not disappeared; its energy sits in its copies.

So the image does keep getting worse with height: the target loses its main lobe and its energy is redistributed. What is bounded is the strength of the copies, and that is what the heatmaps show. If the second view is wanted in the sweeps, it is one extra column (peak of each target

divided into its ambiguity).

2.4 Many targets: the sweep

The heatmap reports the largest copy over all targets and windows, so with more targets it sits closer to the bound. Table 3 compares a Monte-Carlo draw of random heights (phases from $h_{2\pi,i}$, largest $|c(d)|\beta_{|d|}$ over the targets) with the saturated cells of the sweep, and Fig. 4 checks all 1053 cases against Eq. (4) applied target by target: without SATA the whole sweep is explained by the four phases of each target, down to the level where the metric reads the sidelobes of the neighbours.

N	Monte Carlo, median	Monte Carlo, 10–90 %	sweep, ramps with large heights, $\alpha \geq 20°$: median (min, max)
2	−7.6	−10.6 . . . − 5.7	−7.8 (−10.8, −6.6)
3	−6.9	−9.4 . . . − 5.3	−7.3 (−9.0, −5.3)
5	−6.4	−8.3 . . . − 5.1	−7.1 (−9.0, −5.3)
7	−6.1	−7.5 . . . − 4.9	−7.0 (−8.0, −5.2)
9	−5.9	−7.0 . . . − 4.9	−5.4 (−8.0, −5.2)
13	−5.7	−6.6 . . . − 4.9	−5.2 (−6.1, −5.2)

Table 3: Saturated level without SATA [dB re. reference peak]. Largest value over the 1053 cases: −4.93 dB; bound −4.5 dB.

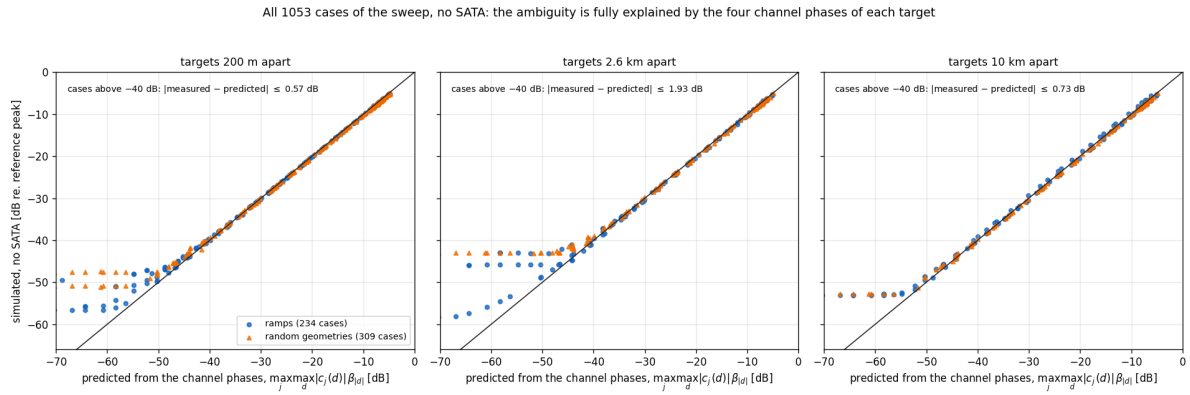


Figure 4: Every case of the sweep without SATA: simulated against predicted from the channel phases of its targets.

2.5 The second plateau: SATA with targets 200 m apart

With SATA the same question arises for the close targets with large height steps, where the level settles at −13 to −14 dB from $\alpha = 20°$ on. The mechanism is different and is worth stating, because it is a property of the map, not of the phase accuracy of SATA.

The map interpolates ΔC_0 linearly between the cells of neighbouring targets (13 cells of 15.4 m for 200 m spacing), while a target's sub-aperture spectrum spans about ±40 cells (the main lobe of the 20-sample triangular window, 50 % of the energy within ±6.5 cells). A target is therefore corrected with a phase that ramps across its own spectrum by the neighbours' step $\Delta\varphi_i(\Delta h)$ per 13 cells (Fig. 5, top). As long as the ramp stays within about ±1 rad over the energy (step $\lesssim$ 200 m) the correction is nearly right. When the step exceeds a cycle on channel 0 ($\Delta h \gtrsim 1$ km, $\Delta\varphi_0 = 303°$ at 946 m), the ramp exceeds 70° per bin: in the sub-aperture FFT a linear phase over the bins is a circular time shift, here of 25 samples against a 20-sample sub-aperture, so the target's energy in that channel is shifted out of the sub-aperture and discarded rather than corrected. Fig. 5 (bottom left) measures it: at $\Delta h = 946$ m channel 0 keeps 7 %

of the target's energy and channel 1 keeps 36 %, while channels 2 and 3 ($h_{2\pi} = 17$ and 42 km) keep 98–100 %. The reconstruction then works with two good channels and two nearly empty ones, $g \approx (0.2, 0.5, 1, 1)$, and Eq. (4) with $e^{j\varphi_i}$ replaced by g_i gives $|c(\pm 1)| \approx |-1 \pm j|/4 = 0.35$, i.e. -13.5 dB after β_1: the plateau. It predicts the simulated level within 0.5 dB at every step (bottom right), and it also costs the main lobe: in the sweep the inner targets of the $N = 5$ ramps with 946–3100 m steps lose 4.8–7.1 dB with SATA (against 0.8 dB at a 227-m step). The plateau does not move with height because the two short baselines would need steps of 17 and 42 km to be affected. Targets 2.6 km apart have their own block in the map and do not see this at all.

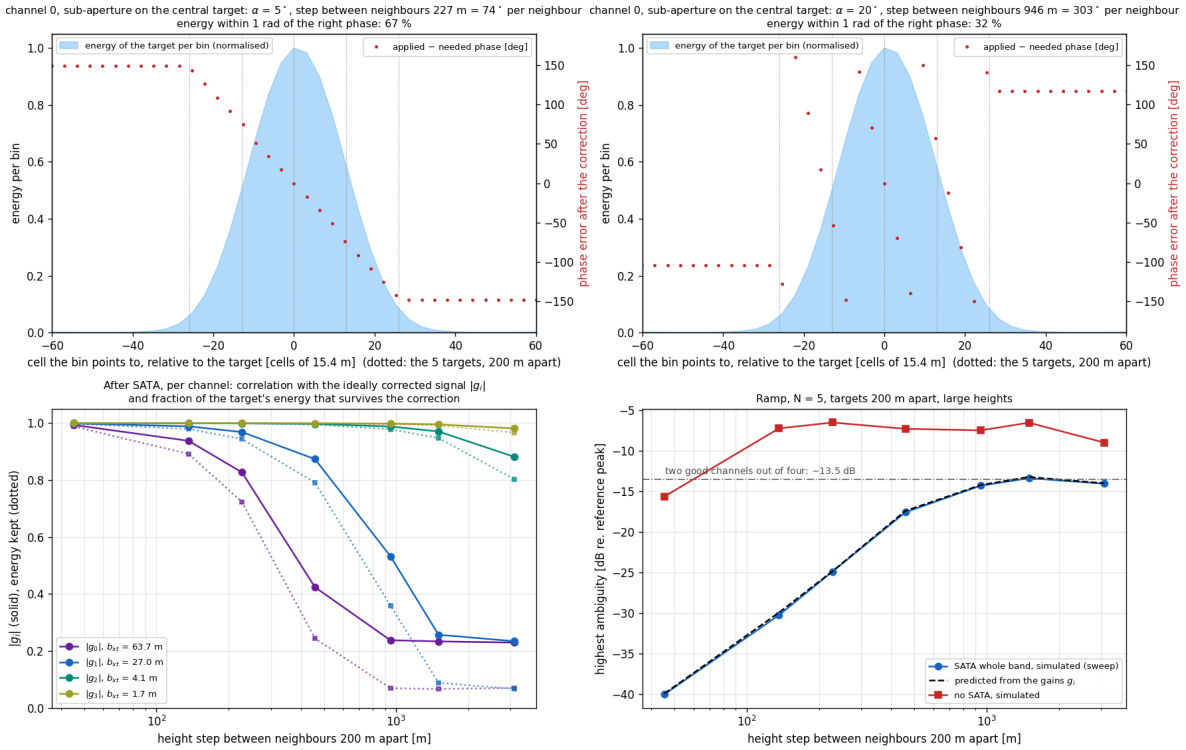


Figure 5: Ramp, $N = 5$, 200 m apart, large heights. Top: energy of the central target per bin of its sub-aperture spectrum and the phase error left after the correction, for a 227-m and a 946-m step. Bottom left: per channel, correlation of the corrected signal with the ideally corrected one and fraction of the energy kept. Bottom right: the plateau and its prediction from the gains.

Conclusion. Without SATA the ambiguity saturates because the reconstruction can only redistribute each target's energy between its main lobe and at most four copies, and the copies are bounded by the fraction of the band they keep; the phases wrap after about 1 km of height on the longest baseline, after which more height changes the numbers without increasing them. Relative to the target's own collapsing peak the ambiguity does keep growing, without bound. With SATA and close targets the plateau at -13.5 dB is set by the two short-baseline channels surviving while the two long-baseline ones are emptied by the steep correction ramp between neighbours. Both plateaus are reproduced by closed-form expressions from the channel phases (or gains) alone, to better than 1 dB.

Scripts (`sata1d_code/probes/`): `deep_q1_residual.py` (Table 1), `deep_q1_doppler_residual.py` (Table 2), `deep_q2_single.py` (Fig. 3), `deep_q2_sata_plateau.py` and `probe_sata_gain.py` (Fig. 5), `plot_deep.py` (figures).